

TEMPLE DONATION AND MANAGEMENT SYSTEM

Scenario-Based Requirement Analysis and Software Requirement Specification

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1. Introduction

The Temple Donation and Management System is a software application designed to manage temple donations, devotees, events, receipts, and administrative activities in an efficient and secure manner. The system replaces manual donation records with centralized digital management. Devotees can register, make donations, view donation history, and receive digital receipts. Temple administrators can manage donations, devotees, events, reports, and system information.

2. Business Domain

The system belongs to the religious and charitable management domain. It supports temple administration by providing digital facilities for managing donations, devotees, religious events, and financial records.

3. Problem Description

Traditional temple donation management may depend on manual registers, paper receipts, spreadsheets, and separate records. This can lead to data duplication, calculation errors, difficulty in tracking donations, delayed receipt generation, and difficulty in preparing reports. The proposed system provides centralized and secure management of donation and temple-related information.

4. Stakeholders

Devotees/Donors; Temple Administrator; Temple Management; Account/Finance Staff; System Developer; Testing Team; Payment Gateway; Email/SMS Notification Service.

5. Existing Challenges

Manual recording of donations; paper-based receipts; difficulty tracking donor history; data duplication and human errors; difficulty generating financial reports; limited real-time information; difficulty managing temple events; security concerns with physical records; time-consuming administrative work.

6. System Objectives

1. Automate temple donation management. 2. Provide secure online donation facilities. 3. Maintain accurate donor and donation records. 4. Generate digital donation receipts. 5. Allow devotees to view their donation history. 6. Manage temple events and announcements. 7. Generate donation and financial reports. 8. Reduce manual work and errors. 9. Provide secure role-based access. 10. Improve overall temple administration.

7. Functional Requirements

- FR1 – User Registration: The system shall allow devotees to create an account using required personal information.
- FR2 – User Login: Registered users shall securely log into the system.
- FR3 – Profile Management: Devotees shall be able to view and update profile information.
- FR4 – Donation Management: Devotees shall be able to select a donation type and enter the required donation amount.
- FR5 – Online Payment: The system shall process online donations through an integrated payment gateway.
- FR6 – Donation Receipt: The system shall generate a digital receipt after successful donation.
- FR7 – Donation History: Devotees shall be able to view previous donation details.
- FR8 – Event Management: Administrators shall be able to add, update, and delete temple events.
- FR9 – Event Viewing: Devotees shall be able to view upcoming temple events and announcements.
- FR10 – Donor Management: Administrators shall be able to view and manage donor information.

- FR11 – Donation Records: Administrators shall be able to view, update, and manage donation records.
- FR12 – Report Generation: The system shall generate donation, donor, event, and financial reports.
- FR13 – Notifications: The system shall send donation confirmations and important event notifications.
- FR14 – User Management: Administrators shall be able to manage user accounts and roles.
- FR15 – Search: Authorized users shall be able to search donation, donor, and event information.

8. Non-Functional Requirements

Performance

The system should provide fast responses during donation, searching, and report generation.

Security

User information and financial data should be protected using authentication, authorization, secure communication, and protected database access.

Reliability

The system should accurately record donations and minimize data loss or system failures.

Usability

The interface should be simple and easy for devotees and administrators to understand.

Scalability

The system should support increasing numbers of devotees, donations, and events.

Maintainability

The system should be easy to modify, test, debug, and maintain.

Availability

The system should remain available whenever devotees and administrators need to access it.

9. Actors and Use Cases

- Devotee: Register, Login, Manage Profile, Make Donation, View Receipt, View Donation History, View Events, Receive Notifications.
- Administrator: Login, Manage Users, Manage Donors, Manage Donations, Manage Events, Generate Reports, Monitor System Activities.
- Temple Management: View Reports, Monitor Donations, Analyze Temple Activities.
- Payment Gateway: Process online payments.
- Email/SMS Service: Send confirmations and notifications.

10. Use Case Summary

Actor	Major Use Cases
Devotee	Register, Login, Profile, Donation, Receipt, History, Events
Administrator	Login, Users, Donations, Events, Reports
Temple Management	Reports, Monitoring
Payment Gateway	Process Payment
Email/SMS Service	Send Notifications

11. Use Case – Make Donation

Actor: Devotee

Precondition: Devotee must be registered and logged in.

Process: Select donation → enter category and amount → validate → payment gateway → record transaction → generate receipt → send confirmation.

Postcondition: Donation is successfully recorded.

12. Use Case – Manage Events

Actor: Administrator

Precondition: Administrator must be logged in.

Process: Open event management → add/update/delete event → enter details → validate → save → update database → display event to devotees.

Postcondition: Temple event is successfully managed.

13. Data Requirements

- Devotee details
- Login credentials
- Profile information
- Donation details
- Donation category
- Payment transaction details
- Digital receipts
- Donation history
- Temple event details
- Announcements
- User roles
- Reports
- System logs

14. External Interface Requirements

- User Interface: Registration, login, dashboard, donation, payment, receipt, history, events, administration, reports.
- Hardware Interface: Computers, laptops, tablets, and smartphones.
- Software Interface: Database, payment gateway, authentication service, and email/SMS service.
- Communication Interface: Secure HTTPS communication.

15. System Features

- User Authentication
- Devotee Profile Management
- Online Donation
- Payment Processing
- Digital Receipt Generation

- Donation History
- Temple Event Management
- Donor Management
- Report Generation
- Notifications
- Role-Based Administration
- Secure Data Storage

16. Assumptions

- Users have basic knowledge of smartphones or computers.
- Users have Internet access.
- Devotees provide accurate information.
- Administrators maintain system data.
- Payment gateway services are available.
- The database is regularly backed up.
- Temple management follows organizational policies.

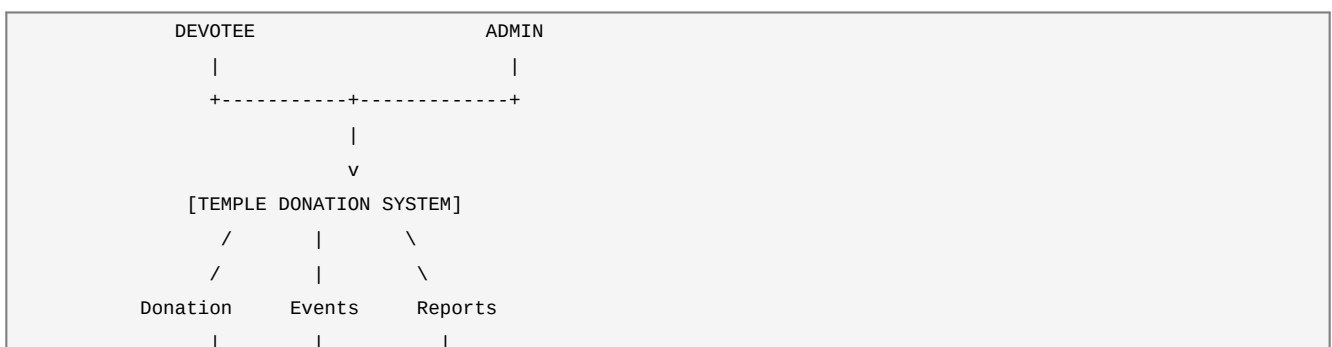
17. Constraints

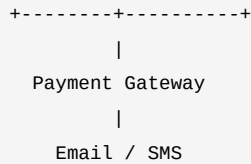
- Technical: Internet dependency, database compatibility, limited server resources.
- Operational: Administrator training and possible maintenance downtime.
- Economic: Development, hosting, and payment-service costs.
- Legal: Protection of personal and financial information.
- Time: Completion within the academic deadline.

18. Requirement Validation

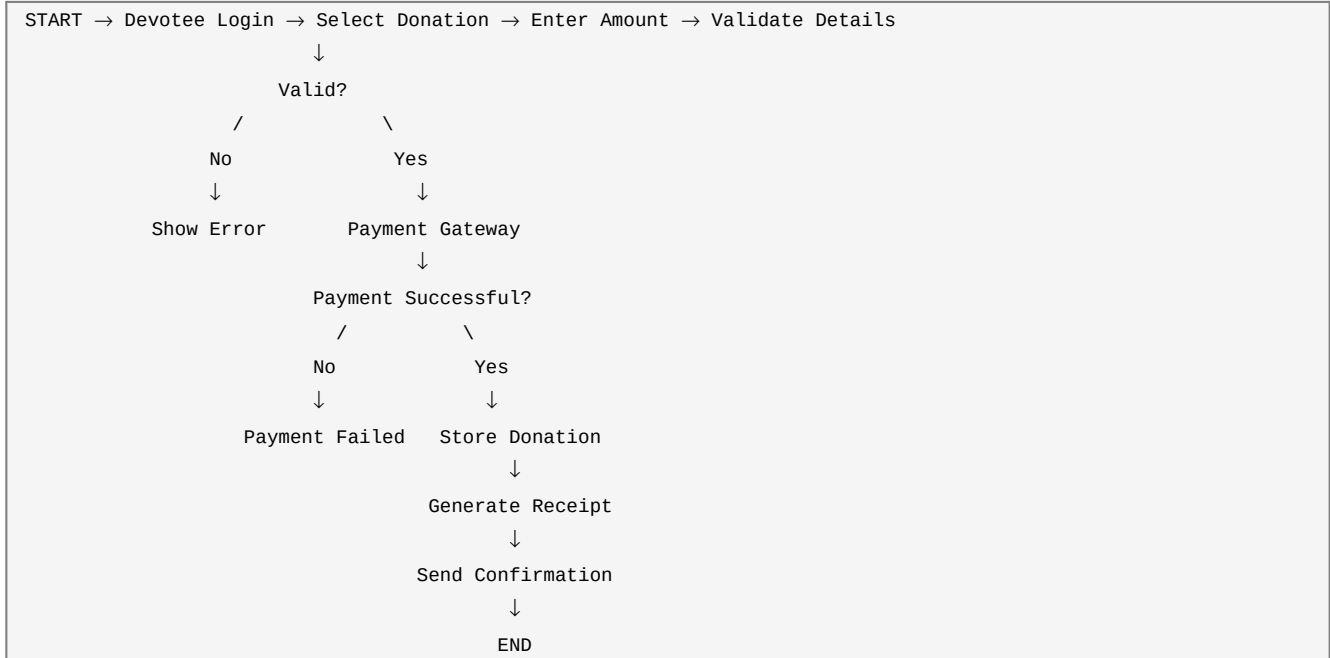
- Completeness: Include all stakeholders, functions, data, and interfaces.
- Consistency: Use cases and functional requirements should not contradict each other.
- Ambiguity: Requirements should be clear and measurable.
- Redundancy: Duplicate requirements should be removed.
- Missing Requirements: Review security, payment failure handling, receipt generation, backup, validation, and error handling.

19. System Context / Use Case Structure

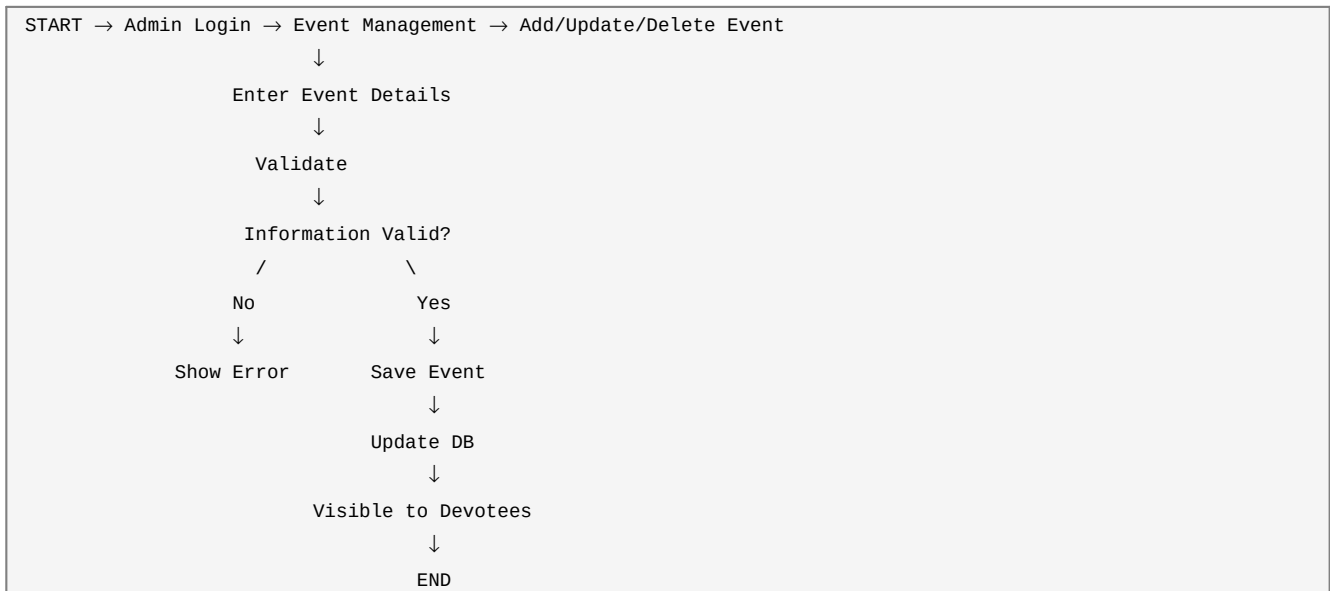




20. Donation Process Flowchart



21. Event Management Flowchart



22. Conclusion

The Temple Donation and Management System provides a centralized solution for managing donations, devotees, events, receipts, and administrative activities. It reduces manual work, improves data accuracy, provides secure donation processing, and enables efficient reporting. The system can be expanded with mobile applications, advanced analytics, multilingual support, and additional temple services.